Worksheet 5

1. When should you use linear search vs. binary search? What do arrays need to have to use binary search?

   Linear search is better-suited for small, unsorted arrays. On the other hand, binary search is helpful when sorting large, sorted arrays; arrays must be sorted in increasing value when using binary search.

2. int[][] arr = {12, 9, 5, 14, 8, 2, 13, 1}
   Sort the array with Selection, Insertion, and Merge Sort.

   a. Selection Sort:

      Pass 0 (original array): {12, 9, 5, 14, 8, 2, 13, 1}
      Pass 1: {1, 9, 5, 14, 8, 2, 13, 12}
      Pass 2: {1, 2, 5, 14, 8, 9, 13, 12}
      Pass 3: {1, 2, 5, 14, 8, 9, 13, 12}
      Pass 4: {1, 2, 5, 8, 14, 9, 13, 12}
      Pass 5: {1, 2, 5, 8, 9, 14, 13, 12}
      Pass 6: {1, 2, 5, 8, 9, 12, 13, 14}
      Pass 7: {1, 2, 5, 8, 9, 12, 13, 14}

   b. Insertion Sort:

      Pass 0 (original array): {12, 9, 5, 14, 8, 2, 13, 1}
      Pass 1: {9, 12, 5, 14, 8, 2, 13, 1}
      Pass 2: {5, 9, 12, 14, 8, 2, 13, 1}
      Pass 3: {5, 9, 12, 14, 8, 2, 13, 1}
      Pass 4: {5, 8, 9, 12, 14, 2, 13, 1}
      Pass 5: {2, 5, 8, 9, 12, 14, 13, 1}
      Pass 6: {2, 5, 8, 9, 12, 13, 14, 1}
      Pass 7: {1, 2, 5, 8, 9, 12, 13, 14}

c. Merge Sort:

Pass 0 (original array): {12, 9, 5, 14, 8, 2, 13, 1}
Pass 1: {9, 12, 5, 14}, {8, 2, 13, 1}
Pass 2: {9, 12}, {5, 14}, {8, 2}, {13, 1}
Pass 3: {5, 9, 12, 14}, {1, 2, 8, 13}
Pass 4: {1, 2, 5, 8, 9, 12, 13, 14}

3. int[][] arr = {3, 6, 9, 13, 20}
   How many comparisons will it take to find the number 20 with binary search?

   9, 13, 20
   3 comparisons

4. int[][] arr = {34, 56, 22, 13, 7, 45}
   Sort the array with Selection, Insertion, and Merge Sort.

   a. Selection Sort:

   Pass 0 (original array): {34, 56, 22, 13, 7, 45}
   Pass 1: {7, 56, 22, 13, 34, 45}
   Pass 2: {7, 13, 22, 56, 34, 45}
   Pass 3: {7, 13, 22, 56, 34, 45}
   Pass 4: {7, 13, 22, 34, 56, 45}
   Pass 5: {7, 13, 22, 34, 45, 56}

   b. Insertion Sort:

   Pass 0 (original array): {34, 56, 22, 13, 7, 45}
   Pass 1: {34, 56, 22, 13, 7, 45}
   Pass 2: {22, 34, 56, 13, 7, 45}
   Pass 3: {13, 22, 34, 56, 7, 45}
   Pass 4: {7, 13, 22, 34, 56, 45}

Pass 5: {7, 13, 22, 34, 45, 56}

c. Merge Sort:

Pass 0 (original array): {34, 56, 22, 13, 7, 45}
Pass 1: {34, 56, 22}, {13, 7, 45}
Pass 2: {34, 56}, {22, 13}, and {7, 45}
Pass 3: {34, 56}, {13, 22}, {7, 45}
Pass 4: {13, 22, 34, 56}, {7, 45}
Pass 5: {7, 13, 22, 34, 45, 56}